Resilient decentralized communication for unstructured peer-to-peer networks

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Abstract

Recent crises, natural disasters, and large-scale protests emphasize how people rely heavily on online social networks and instant messaging platforms to communicate, while organizations use those platforms to coordinate aid efforts. During crises, consolidated communication systems can become strained, partially or wholly destroyed, censored, or otherwise monitored. Free distributed alternatives to consolidated networks are peer-to-peer networks, which distribute the power and content control over many, equal peers. Previous studies evaluated algorithms, metrics and performance of structured overlays of p2p networks. This paper evaluates decentralization metrics of unstructured p2p networks, by providing empirical and simulated insights to their performance and resilience. We propose design considerations to support resilient overlay communication amid clusterization.

1 Introduction

The decentralized administration of the Internet is one of the basic enablers of the rapid growth of the Internet as a whole, thanks to the autonomous nature of different network systems [14]. Different Internet Service Providers (ISPs) play a critical role in the functionality of the Internet, as they connect their end customers (Internet users) to the Internet backhaul. The power dynamic of this system inherently gives control over shaping consumption patterns to the ISPs: consolidation of power at the hands of a few ISPs gives them the unique ability to control, throttle, or monitor the flow of information, in addition to enabling oppressive practices such as censorship and surveillance of end users. Governmental control of the telecommunication infrastructure often leads to the repression of communities during times of political unrest [2, 8], severely limiting the possibility for social actors to dissent and organize themselves. In addition to policial repression engaged by the government, physical distruptions can also limit free speech. Most recently, communication disruptions due to airstrikes on main fiber optic lines, and the depletion of fuel for generators, have severely limited the information flow to the population of Gaza since the start of the Israel-Hamas war in October 2023, halting humanitarian coordination, and leaving the affected population entirely cut off from life-saving information and assistance [7, 26].

Together, the aforementioned practices have led to a growing interest in peer-to-peer (p2p) messaging platforms, decentralized social networks [3, 25], and distributed ledger-based applications [6, 21]. To guarantee free speech in disaster zones, during protests, and for citizens living under oppressive regimes, we decide to focus on unstructured p2p networks, as they inherently exhibit a higher level of resilience to topology changes and churn, in comparison to structured p2p networks. The commonality to most structured p2p networks is the usage of Distributed Hash Tables (DHTs) to guide the communication and storage distribution in the network. Additionally, structured p2p networks usually require an expensive randomized approach to establish resilient topologies, as they need to spread information across random peers to prevent dependence on any single peer, and maintaining the costly needed relations, which constantly deplete, to iteratively diffuse the information incurs latency dips [10–12, 20]. Popular structured p2p networks include systems as Chord [28], Kademlia [20], and the InterPlanetary File System (IPFS) [4]. On the contrary, unstructured p2p networks are more suitable for highly-dynamic alternative communication methods, often necessary to guarantee free decentralized communication means. Most notably, the rise of unstructured p2p messaging platforms like Firechat [27], Briar [24], and Amigo [15], have proven that there is a growing shift in user interest patterns away from traditional Centralized Social Networks towards diverse p2p social networks.

After exploring the different types of p2p overlay networks, we discuss the choice of the communication protocol, which dictates the limited physical device communication radius. We chose to focus on a stable protocol that most modern devices support and the populace is most likely to use in disaster zones: Bluetooth, and more specifically Bluetooth Low Energy (BLE) [5]. By focusing on BLE-based p2p communication, we can conserve device battery life, to allow for critical communication in extreme situations: it allows users to exchange messages over short ranges over a longer period of time than comparable communication protocols e.g. WiFi Direct, which would drain the battery faster in constrained devices [9].

This paper provides the design space for resilient unstructured p2p networks, amid differently sparse underlying network topologies.

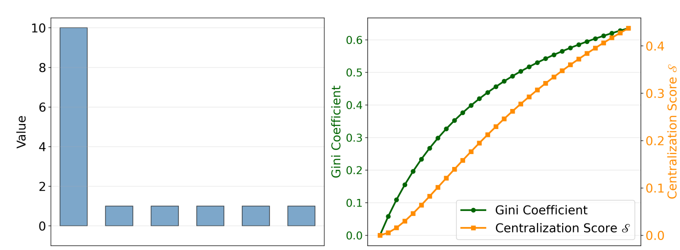

Figure 1: Sparse topology

Figure 1 illustrates an increasingly sparse network topology, in which one node becomes increasingly dominant as it acumulates more network links. While the left plot simply shows the distribution of values, the right plot compares the different resilience scores that this paper uses: the Gini coefficient in green, and the Centralization score $\mathcal{S}$ in orange.

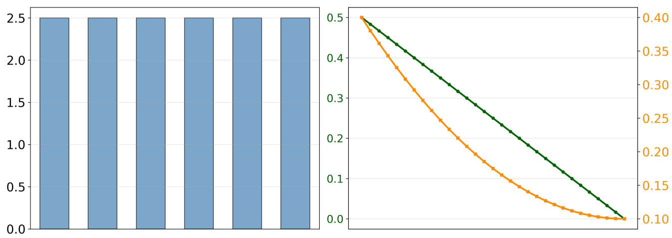

Figure 2: Dense topology

An increasingly dense topology can be observed in figure 2, that illustrates the change in resilience metrics as the topology becomes equally distributed. The value at the start of this plot represents an extremely unequal distribution, where one node dominates all resources (i.e. network links), and as the plot progresses on the X axis, the topology gradually reaches a state of perfect equilibrium where all nodes are equally connected. It is noteworthy to observe that the Gini coefficient at the end of this plot reaches a minimum value of 0, indicating perfect equality, where the $\mathcal{S}$ score does not; this is because the Gini coefficient is calculated by computing the difference in individual concentration of value in the network, and when all those differences are zero, the Gini coefficient also reaches zero. This is because the Gini measurement focuses on the **overall** inequality distribution, where the $\mathcal{S}$ score focuses on the **top** dominant values, which makes it inherently sensitive to outliers.

In this research paper, we conduct a comprehensive representational experiment to collect empirical data on the stochastics of the BLE signal propagation over different distances, and use the collected real-world empirical evidence to parameterize simulations of unstructured p2p networks, and then measure their resilience [13, 22], and theoretical channel capacity. This contribution proposes a resilience metric for local p2p overlays, which captures resilience based on sparsity. For this, in section 2, we discuss the Bluetooth protocol, and usual metrics that capture resilience and centralization. In section 2 we discuss the Bluetooth protocol background, the state of the art of sparsity and resilience metrics as measurable in network topologies, and present related works. Section 3 describes the setup of the experiment testbed that captures the BLE measurements, presents the empirical results from the testbed which we then use as parameters for the simulations, and closes with detailing the simulation setup 3.4. Following the setup section, section 4 evaluates the results from the simulations, concluded by the discussion on section 5.

2 State of the Art

To provide decentralized p2p communication, Lindgren et al. move away from traditional IP packet forwarding and propose the the Store-Carry-Forward paradigm using Delay-Tolerant Networking (DTN) [19], where hosts store messages while carrying them until another another node is reached which can then forward the messages. This paradigm enables the opportunistic usage of technologies, such as Wi-Fi Direct or Bluetooth (ubiquitous to virtually all smartphones and computers), to forward messages while the Internet infrastructure is unavailable, as Legendre et al. present with Twimight [18]. Twimight and most unstructured peer-to-peer networks usually rely on flooding, efficient clique-level forwarding, and gossip for content dissemination, as recently exemplified with Amigo [15] and bitchat [16], as this type of networks typically do not store information about the cluster membership. The core principles of this research paper are the concepts of flooding and clique-forwarding, as we measure the resilience and performance metrics for unstructured p2p social networks that are built using flooding and clique-forwarding, particularly important for speedy neighbor discovery and content dissemination. The resilience and performance metrics of different common Internet services vary wildly between different countries, depending on the centralization of those services, as centralization of services directly impacts their host countries' overall capacity to mitigate outages, and minimize content delivery latency for end-customers. In [13] Habib et al. study patterns of different web service providers in different countries around the world. Habib demonstrates how countries with higher centralization around a smaller number of website hosting providers are more susceptible to outages, and propose a metric called the Centralization Score $\mathcal{S}$, which Habib uses to measure the centralization of common Internet services such as web hosting, DNS, and CAs around a small number of providers. The centralization Score $\mathcal{S}$ is derived from the Herfindahl-Hirschmann Index (HHI), which is used to measure market competition in US antitrust laws. The property we used to measure resilience is the node degree, or the number of connections a node has to other peers in the network topology. An overall similarity between individual node degrees increases the resilience of a topology, as network partitions affecting a subset of the nodes will not result in a complete blackout of the network because many alternative paths still exist. The Centralization Score $\mathcal{S}$ is defined as follows:

$$\mathcal{S} = \sum_{i=0}^{n} (\frac{a_i}{C})^2 - \frac{1}{C}$$

where C is the total number of websites considered, a_i is the number of websites hosted by provider i, and n is the total number of providers. In this paper, C is defined as the total number of edges (i.e. network links), a_i is the degree of node i, and n is the total number of nodes. To sparsity: in their work, Hurley and Rickard

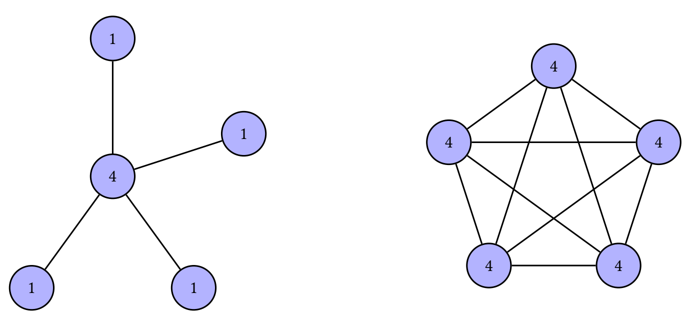

Figure 3: Sparse vs. dense topologies

compare different measures of sparsity [22], and rank different known sparsity measures according to six desirable criteria that a sparsity measure should trivially satisfy, most notably the Robin Hood axiom ('stealing from the rich and giving to the poor': distributing the website hosting over multiple providers instead of hosting with single provider should reduce sparsity). Hurley et al. prove that amongst the most common methods to measure sparsity, the Gini coefficient outperformed all other measurements, and satisfies all six criteria. The Gini coefficient, a metric commonly associated with the Human Development Index to measure income inequality around the world, is calculated as follows:

$$G = \frac{\sum_{i=0}^{n} \sum_{j=0}^{n} |a_i - a_j|}{2n \sum_{i=0}^{n} a_i}$$

Where a_i is the degree of node i, and n is the total number of nodes. Hurley and Rickard's work allows us to effectively use tried-and-tested economical metrics to measure the sparsity of the different simulation topologies. Finally, this paper compares the aforementioned metrics with the trivial L0 norm, which is calculated from the different topologies' node degrees followingly:

$$L_0 = \frac{\sum_{i=0}^{n} \begin{cases} 1 & \text{if } a_i > 0 \\ 0 & \text{otherwise} \end{cases}}{n}$$

where a_i is the node degree of node i, and n is the total number of nodes.

3 Setup

3.1 Testbed

Originally we intended to measure WiFi signal propagation in addition to Bluetooth, we decided to focus on the latter as it is more power efficient, and requires no network infrastructure. Furthermore, Bluetooth Low Energy (BLE) was picked over classic Basic Rate/Enhanced Data Rate (BR/EDR), as BLE is even more efficient for (small) message exchanges [5]. To measure the BLE signal coverage stochastics and how the signal propagates in free space, we compared the Received Signal Strength Indicator (RSSI) of BLE advertisements at the receiver node across increasing distances from the transmitter node. BLE ads form the bottleneck of BLE communications, as ads are used to discover and connect to neighbours in local networks. Due to the half-duplex nature of BLE, peers are not able to advertise while they are scanning for BLE ads. Two types of scanning are possible: passive scanning, where the scanner only listens for ads, and active scanning, where upon receiving an ad, the scanner also sends extended scan requests to the advertiser node for additional data. In this experiment, passive scanning is used, because it is sufficient to discover neighbors and characterise the RSSI, while active scanning could affect the capacity of the advertising pi to send ads, as it would have to answer the extended scan requests. Our testbed consists of a set of Raspberry Pi 3B+ devices, using the built-in BLE supporting chip Cypress CYW43455, and external power supplies of 5v/3.4A to ensure stable operation and mobility. The Raspberry Pis were flashed with a lightweight 64-bit Linux port (Raspberry Pi OS Lite based on Debian Trixie), upon which the basic Bluetooth stack was installed using BlueZ 5.66.

3.2 Empirical Insights

During the experiment, one Raspberry Pi acted as the transmitting node, using an automated bluetoothctl script to continuously advertise at a fixed interval of 20ms $+\mathcal{X}$, where $\mathcal{X}$ is a random variable uniformly distributed between 0 and 10ms, representing the delay added to avoid collisions with other devices using the same advertising interval [5, 23]. Another Raspberry Pi performed the receiver role, running a Python script that uses the Bleak library to continuously scan for a window set to one minute. The two devices were placed at increasing distances from each other, where the advertiser node was kept static and the receiver node was moved. At every distance step, the receiver node recorded the RSSI of all ads received from the transmitting node, filtering them by the unique address of the advertising node, and logging the RSSI values, as well as the transmitting power (tx) and a timestamp, to a file. Each distance step was repeated twenty times to capture enough data points for statistical significance. The distances measured were from 0 meters to 10 meters in increments of 5 meter, and then from 10 meters in increments of 10 meters, until no more ads were received for most runs of that distance. The experiment was divided to two parts: in the first part, the distance points up to 30 meters were evaluated in an urban backyard; in the second part, the distance points from 40 meters were evaluated in an olympic athletics field. Both parts of the experiment were conducted in an open level environment with minimal surrounding interference, clear direct line of sight between the nodes, and minimal multipath effects.

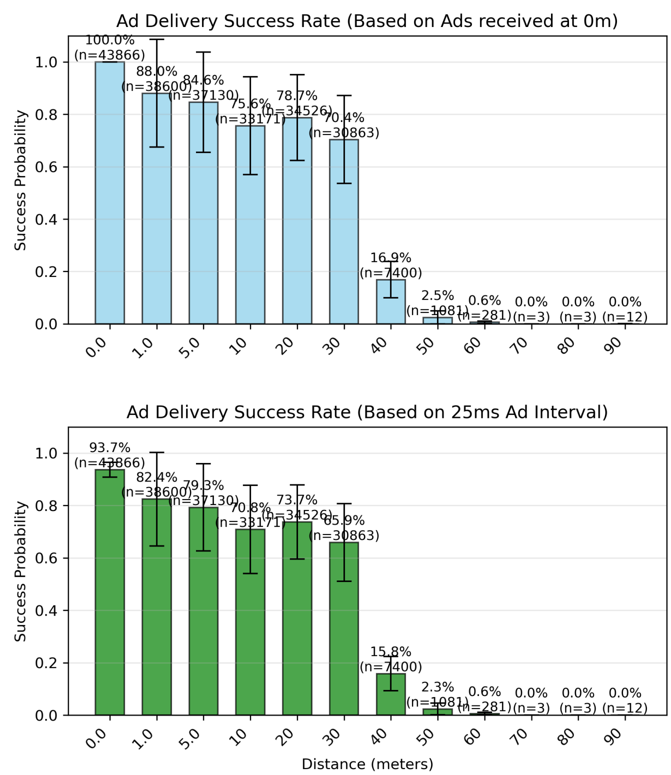

Figure 4: Advertisement Delivery Success Probability

Figure 4 shows the advertisement delivery success probability at different distances, or the percentage of ads received at the receiver node, depending on two estimates of the number of ads transmitted. The figure showcases two subplots: the top plot calculates the ratio between the count of ads received at distance X, and the count of ads received at 0 meters, where the two devices were placed right next to each other. This plot assumes that the count of ads received at 0 meters is the same as the count of ads that were actually sent. The bottom plot estimates the number of ads that the transmitter node sent, by dividing the length of time of the scanning interval by the advertising interval of 20ms and adding to it the expected value of the random variable $\mathcal{X}$: 5ms. It is trivial to see that as the distance increases, the advertisement delivery success probability decreases, following a roughly exponential decay. No ads were received for any run for distances more than 100 meters, so the experiment was concluded at that point.

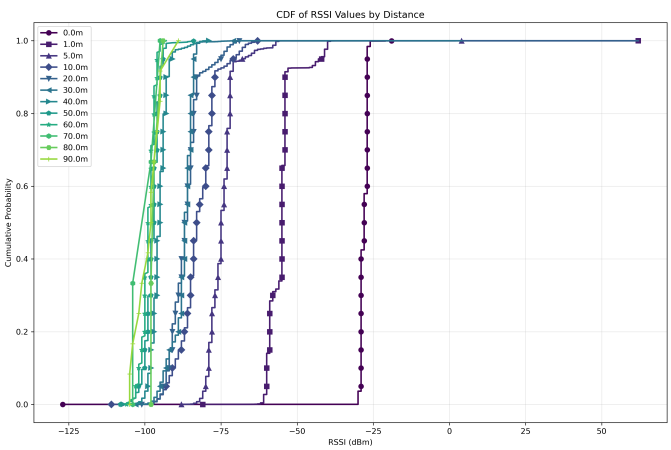

Figure 5: RSSI CDF

The Cumulative Distribution Function (CDF) plot of the RSSI values received at different distances is shown in figure 5. As expected, the figure shows a general decreasing trend of the RSSI values, in correlation with the increase in distance. The jittery nature of the CDF curves at distances ≥ 40 meters can be attributed to the lower number of samples collected at those ranges, which leads to a higher variability in the RSSI measurements. The Probability Density Function (PDF) plot and a time series plot of the RSSI values for each distance can be found in the appendix, figures 10 and 11.

Next, table 1 shows summary statistics of the RSSI values received at different distances: the mean RSSI values presented at the table were used to parameterize the simulation signal propagation decay with distance. Throughout all distances, the same transmitting power was recorded and thus not displayed in the table, which hints at the default transmitting power of a Raspberry Pi 3b+: 12 dBm.

Table 1: RSSI Statistics by Distance

Distance (m)	Mean RSSI (dBm)	Std Dev	Num Samples
0.0	−28.03	1.21	43,866
1.0	−55.14	4.62	38,600
5.0	−74.58	4.77	37,130
10.0	−82.47	6.21	33,171
20.0	−86.54	5.05	34,526
30.0	−87.45	3.55	30,863
40.0	−95.21	2.76	7,400
50.0	−97.50	2.44	1,081
60.0	−98.82	2.10	281
70.0	−99.00	3.74	3
80.0	−96.33	1.70	3
90.0	−98.25	4.23	12

3.3 Simulations

We used the The ONE simulator [17] to simulate an opportunistic exchange of messages between nodes communicating over the BLE, using the empirical results from the testbed experiment to model the signal path loss with distance. The simulation was repeated for 25 runs per configuration, using an Ubuntu 22.04 with an Intel(R) Xeon(R) Gold 6326 CPU @ 2.90GHz with 64 cores, and 247 GB of RAM. To run the simulations we used parallelization, spawning 60 parallel jobs. Using a static movement model, every node remains static (i.e. the node does not move) and attempts to send the same amount of messages to all other nodes in the network. In a disaster zone, it is critical to disseminate information as fast and as far as possible, and due to instability, no routes can be assumed to exist. To mimic this scenario of message propagation in disaster zones, we used an epidemic router to flood messages across the network. In an epidemic routing scheme, every node sends all messages, including messages it relays, to all of its neighbors, and each neighbor recursively repeats this iterative process until the final destination node for each message is reached. For every simulator run, we used either exclusively intra-cluster communication to simulate clique-based short-range cluster communication, or inter-cluster communication to simulate long-range communication: In the former, a host generates 20 messages to all other hosts in its cluster, while at latter every host generates 20 messages to all other hosts in the network. We used the typical BLE MTU size of 247 bytes for all messages. A simulator run ends when all generated messages are either delivered or discarded. For the signal path loss model, we used empirical data presented in table 1, and interpolated for distances between the empirical measurements using the normal logarithmic path loss model laid out in equation 6 of [1]. The path loss model is defined as follows:

$$RSSI_d(dBm) = A - 10 * \alpha * \log_{10}(\frac{d}{d_0})$$

where $RSSI_d$ is the received signal strength indicator at distance d between source and destination nodes, d_0 is the distance at the nearest lower or equal reference point, A is the RSSI at the nearest lower or equal reference distance d_0 (as measured by the testbed), and α is the path loss exponent. We wanted to evaluate the performance of the Bluetooth Low Energy running on 1M PHY mode, which means a modulation rate of 1Mbps at ≤ 1 meter. To calculate the possible channel bitrate cap at a specific range, we used the Shannon-Hartley theorem to derive the theoretical capacity based on known BLE channel properties. A large 400×400 meter L-shape building hall was generated, split to a grid of 12 equal-sized cells of 100 × 100 meters each assigned to a cluster consisting of six hosts, for a total of 72 hosts. We randomly placed each host within the cell to which its cluster was assigned. To generate the random node positions, a random number generator was used. We fed the random generator with a seed that was mathematically derived from the communication range, message size, and optionally by the simulation run number, to ensure result reproducibility. The communication range is a configurable parameter which limits the radius of communication of the network interface that each node has: for simplification, a connection between two nodes can be established if they are within the communication radius of each other, and have direct line of sight. A connection does not churn during a simulation run: once a connection is established it is maintained until the end of the simulation run.

3.4 Model Insights

We used the communication radii of [1,5,10,20,30,40,50,60,70,80,90,100] meters to generate different topologies. A simulation run configuration consists of a set of parameters, including the message generation rate, the maximum communication radius of each node, the maximum number of parallel connections (max node degree) each node can establish and maintain, and the communication mode. The same exact topology generated for each communication radius was repeated for [1,2,3,4,5,6,7,8,9,10,15,20] maximum node degrees, each for 25 runs, keeping the topologies invariant to the run number. In doing so, this shuffled the order of the hosts picked for message generation, which we used to prevent introducing bias by the order of the hosts. This was done to elicit the effects of the maximum node degree and communication range on timely message delivery.

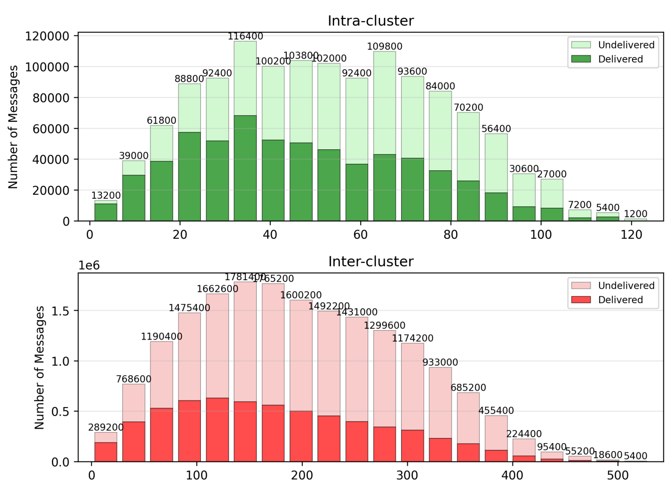

Figure 6: Message Distance Distribution

Figure 6 shows the distance distribution of all messages created, aggregated over all simulation runs, communication ranges, and maximum node degrees tested, split by the communication mode. It is evident that the intra-cluster communication mode had a generally higher overall delivery probability, possibly attributed to a lesser strain on the individual hosts' message buffers.

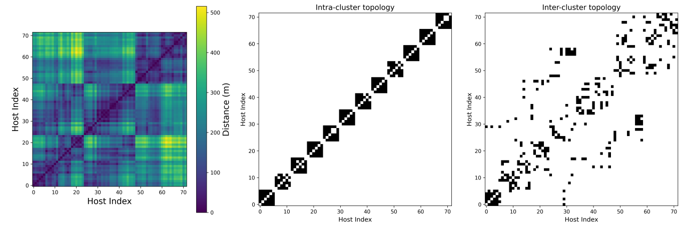

Figure 7: Topology and distance metrics for communication radius 100m, constrained maximum node degree 5

To better understand the topology, figure 7 shows the distance matrix between hosts to the left, supplemented by two topological matrices, one for each communication mode. The topological matrices have a black cell at position (i, j) if host i is connected to host j. Next, randomized topologies were created for each configuration, to assess how the maximum node degree affects the message latency, centralization, and sparsity metrics of unstructured p2p networks. This was done to evaluate how different topologies with the same communication radius and maximum node degree induce variance in the different metrics

4 Evaluation

Finally, to evaluate the performance of different topologies by their centralization metrics, we use the Gini and Centralization Score $\mathcal{S}$ and plot them against the maximum constrained node degree. Figure 8 shows how centralization correlates with mean message delivery latency in: in this plot, a clear relationship is visible for most communication radii between an increase in the communication scores, and an increase in the end-to-end message latency. The exception lies with the 100m communication radius, which exhibits a negative correlation between centralization and latency: the latency decreases with an increase in centralization. This could be attributed to the fact that nodes in this communication radius was able to establish the all links as most nodes cover more than one cluster cell, which enables the nodes to reach a target node with less hops. Here, the $\mathcal{S}$ Score is more sensitive to the outliers in the higher latency ranges, which indicates it is a better metric to capture the impact of centralization on message latency, than the Gini coefficient.

Figure 9 shows how the different centralization metrics introduced in section 2 correlate with an increase in the maximum constrained node degree. An extended version of this plot, comparing the aforementioned centralization metrics to the L0 norm also in the intra-cluster communication mode, can be found in the appendix, figure 13. As evident from this plot, the Gini coefficient has a higher variance in comparison to the $\mathcal{S}$ score, which appears to be more stable as the maximum constrained node degree increases. This indicates that the Gini coefficeint is more appropriate to measure resilience.

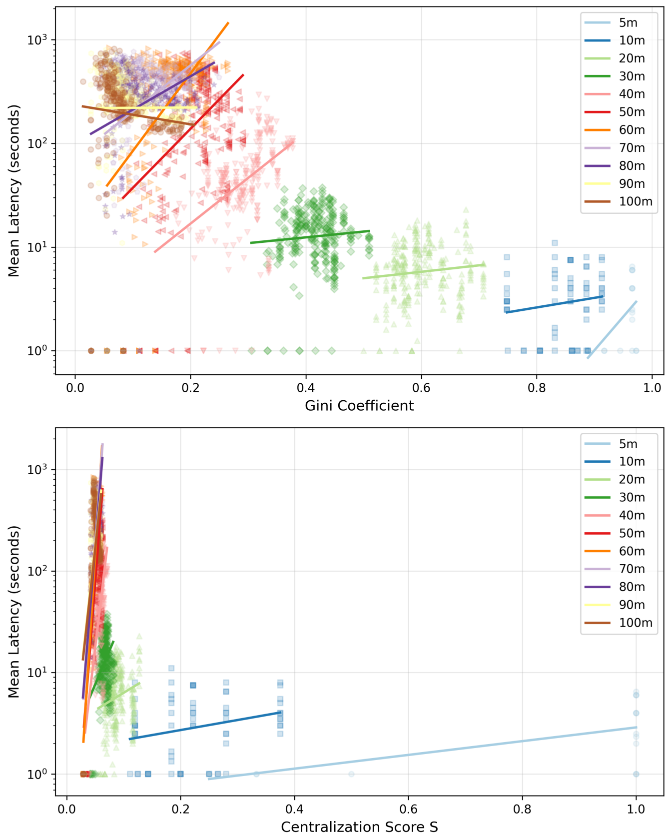

Figure 8: Latency vs. Centralization - inter-cluster communication

5 Discussion

The results shown at figure 9 and in more depth at figure 13 in the appendix, indicate that increasing the maximum constrained node degree generally increases the inequality and centralization of the resulting topology, as measured by both the Gini index and the Centralization Score $\mathcal{S}$. This effect is more pronounced in higher communication ranges, as visible with the stagnation in the $\mathcal{S}$-score for medium communication ranges 50-80m, around the maximum allowed node degree of 10. The Gini coefficient exhibits less of a stagnation effect, which indicates it is more sensitive to changes in the topology than the $\mathcal{S}$-Score. This is due to the fact that the Gini coefficient depends on the entire node degree distribution, whereas the $\mathcal{S}$-Score depends on the number of links and the distribution of degrees, which leads to the increase in number of links cancelling out the increase in node degrees. The Gini coefficient shows a more consistent and steady increase in correlation with an increase in the maximum allowed node degree. To design a topology that is more resilience, it is therefore important to consider both the intended communinication range, and the physical distribution of nodes: if the nodes are geographically dispersed and a higher communication range is needed, the established network might result in a sparsely-populated network which is subsceptible to centralization effects such as vulnerability to outages.

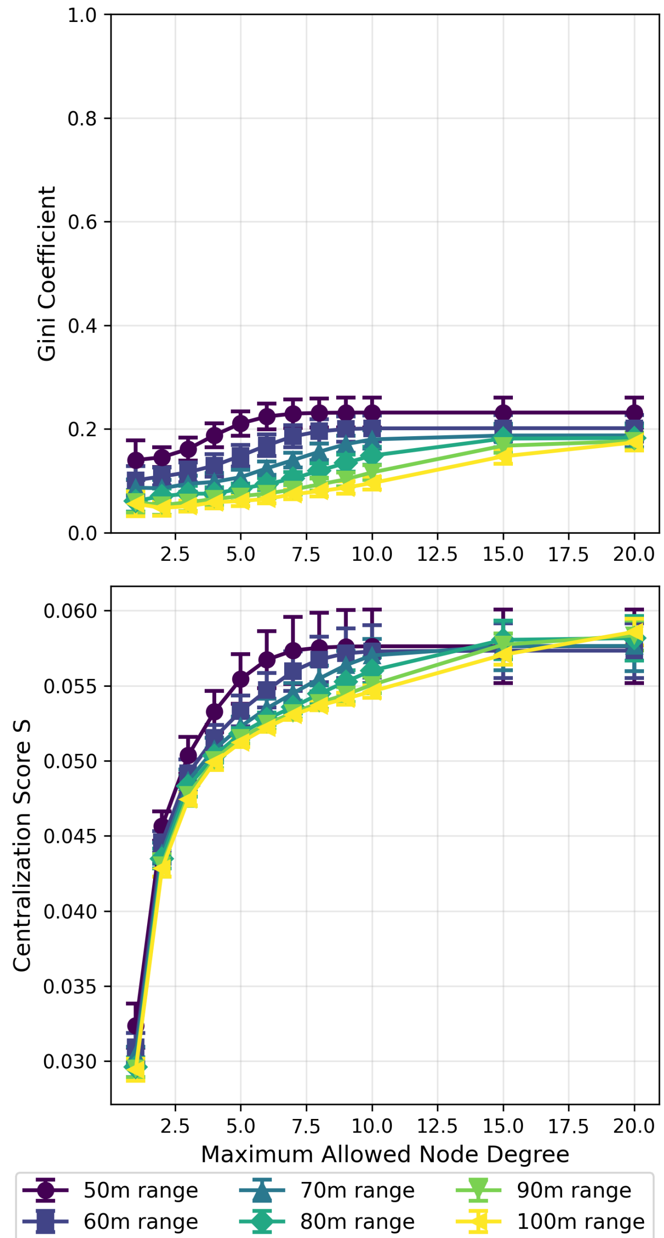

Figure 9: Centralization vs. max node degree - inter-cluster communication

6 Future Work

Due to the inherent variance in the BLE protocol and high interference demonstrated on wireless channels commonly used in the urban settings used for the empirical experiment, some RSSI values should be collected again, as visible from jittery subplots in the PDF plot 10: for example, the 60m and 90m plot exhibit repeating jumps in their density values, in comparison to the smoothness in lower ranges (e.g. 10, 20, 30m ranges). This we attribute to the low number of received packets at those distances (sometimes below 100 packets 4), which makes the RSSI distribution more sensitive to outliers. The RSSI values we have provided for the ranges that are complete and stable, are mostly sufficient to derive the necessary path loss parameters for the simulation, as they exhibit the expected power-law distribution and are mostly consistent. Furthermore, we were not able to have fine-grain control over exactly when the advertisements were sent, because the BlueZ Bluetooth communication stack and the underlying hardware controller handle the advertisement scheduling internally to prevent antenna overheating. To best mitigate this, we set the advertisement interval window to the minimum possible values of to achieve the highest consistent frequency of advertisements sent. We were however able to prove that the advertisements were not actually being consistently received at the configured interval window of 20ms using a Bluetooth sniffer on a mobile phone, which measured the advertisement intervals.

References

[1] Farid Yuli Martin Adiyatma, Dwi Joko Suroso, and Panarat Cherntanomwong. 2023. Regression-based Path Loss Model Correction to Construct Fingerprint Database for Indoor Localization. In *Proceedings of the 2023 6th International Conference on Electronics, Communications and Control Engineering* (Fukuoka, Japan) *(ICECC '23)*. Association for Computing Machinery, New York, NY, USA, 181–186. doi:10.1145/3592307.3592335

[2] Ghadeer Awwad and Kentaro Toyama. 2024. Digital Repression in Palestine. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 393, 15 pages. doi:10.1145/3613904.3642422

[3] Leonhard Balduf, Saidu Sokoto, Onur Ascigil, Gareth Tyson, Björn Scheuermann, Maciej Korczyński, Ignacio Castro, and Michaundefined Król. 2024. Looking AT the Blue Skies of Bluesky. In *Proceedings of the 2024 ACM on Internet Measurement Conference* (Madrid, Spain) *(IMC '24)*. Association for Computing Machinery, New York, NY, USA, 76–91. doi:10.1145/3646547.3688407

[4] Juan Benet. 2014. IPFS - Content Addressed, Versioned, P2P File System. arXiv:1407.3561 [cs.NI] https://arxiv.org/abs/1407.3561

[5] Bluetooth Special Interest Group. [n. d.]. Bluetooth Core Specification Version 4.2. https://www.bluetooth.com/specifications/specs/core-specification-4-2/

[6] Vitalik Buterin et al. 2013. Ethereum white paper. *GitHub repository* 1, 22-23 (2013), 5–7. https://ethereum.org/content/whitepaper/whitepaper-pdf/Ethereum_Whitepaper_-_Buterin_2014.pdf

[7] Emergency Telecommunications Cluster. 2023. Communication and Information Dynamics in Gaza. Brief. Available at: https://www.etcluster.org/sites/default/files/documents/Communications_and_Information_Dynamics_in_Gaza_December_2023_1.pdf. Accessed on: 2025-11-03 21:00 GMT+0200.

[8] Association for Progressive Communications Derechos Digitales. 2022. Internet shutdowns and human rights. OCHR Submission. Available at: https://www.apc.org/sites/default/files/internet_shutdowns_and_human_rights_ohchr_submission_2022.pdf. Accessed on: 2025-11-03 21:00 GMT+0200.

[9] Roy Friedman, Alex Kogan, and Yevgeny Krivolapov. 2013. On Power and Throughput Tradeoffs of WiFi and Bluetooth in Smartphones. *IEEE Transactions on Mobile Computing* 12, 7 (2013), 1363–1376. doi:10.1109/TMC.2012.117

[10] David Guzman, Dirk Trossen, Mike McBride, and Xinxin Fan. 2022. Insights on Impact of Distributed Ledgers on Provider Networks. In *Blockchain – ICBC 2022: 5th International Conference, Held as Part of the Services Conference Federation, SCF 2022, Honolulu, HI, USA, December 10–14, 2022, Proceedings* (Honolulu, HI, USA). Springer-Verlag, Berlin, Heidelberg, 3–17. doi:10.1007/978-3-031-23495-8_1

[11] David Guzman, Dirk Trossen, and Joerg Ott. 2023. If Iterative Diffusion Is The Answer, What Was The Question?. In *Proceedings of the 2nd ACM SIGCOMM Workshop on Future of Internet Routing & Addressing* (New York, NY, USA) *(FIRA '23)*. Association for Computing Machinery, New York, NY, USA, 29–34. doi:10.1145/3607504.3609288

[12] David Guzman, Dirk Trossen, and Joerg Ott. 2025. *Multicast Applicability for Distributed Consensus Systems*. Internet-Draft draft-guzman-multicast-applicability-dcs-00. Internet Engineering Task Force. Work in Progress, Expired 4 September 2025.

[13] Rumaisa Habib, Kimberly Ruth, Gautam Akiwate, and Zakir Durumeric. 2025. Formalizing Dependence of Web Infrastructure. In *Proceedings of the ACM SIGCOMM 2025 Conference* (São Francisco Convent, Coimbra, Portugal) *(SIGCOMM '25)*. Association for Computing Machinery, New York, NY, USA, 1132–1153. doi:10.1145/3718958.3750492

[14] Mark Handley. 2006. Why the internet only just works. *BT Technology Journal* 24, 3 (2006), 119–129. doi:10.1007/s10550-006-0084-z

[15] David Inyangson, Sarah Radway, Tushar M. Jois, Nelly Fazio, and James Mickens. 2025. Amigo: Secure Group Mesh Messaging in Realistic Protest Scenarios. In *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security* (Taipei, Taiwan) *(CCS '25)*. Association for Computing Machinery, New York, NY, USA. doi:10.1145/3719027.3765133 Preprint available at https://eprint.iacr.org/2024/1872. Accessed on: 2025-11-03 21:00 GMT+0200.

[16] multiple open source contributors Jack Dorsey. 2025. bitchat. GitHub. Available at: https://github.com/permissionlesstech/bitchat/blob/main/WHITEPAPER.md. Accessed on: 2025-11-03 21:00 GMT+0200.

[17] Keränen, Ari and Ott, Jörg and Kärkkäinen, Teemu. 2009. The ONE simulator for DTN protocol evaluation. In *Proceedings of the 2nd International Conference on Simulation Tools and Techniques* (Rome, Italy) *(Simutools '09)*. ICST (Institute for Computer Sciences, Social-Informatics and Telecommunications Engineering), Brussels, BEL, Article 55, 10 pages. doi:10.4108/ICST.SIMUTOOLS2009.5674

[18] Franck Legendre. 2011. 30 years of ad hoc networking research: what about humanitarian and disaster relief solutions? what are we still missing?. In *Proceedings of the 1st International Conference on Wireless Technologies for Humanitarian Relief* (Amritapuri, Kollam, Kerala, India) *(ACWR '11)*. Association for Computing Machinery, New York, NY, USA, 217. doi:10.1145/2185216.2185279

[19] Anders Lindgren and Avri Doria. 2007. Experiences from deploying a real-life DTN system. In *Proceedings of the Fourth IEEE Consumer Communications and Networking Conference*. IEEE, Las Vegas, Nevada, USA, 217–221. doi:10.1109/ccnc.2007.50

[20] Petar Maymounkov and David Mazières. 2002. Kademlia: A Peer-to-Peer Information System Based on the XOR Metric. In *Revised Papers from the First International Workshop on Peer-to-Peer Systems (IPTPS '01)*. Springer-Verlag, Berlin, Heidelberg, 53–65.

[21] Satoshi Nakamoto. 2008. Bitcoin: A peer-to-peer electronic cash system. *Available at SSRN 3440802* (2008).

[22] Nick Hurley, Scott Rickard. 2008. Comparing measures of sparsity. In *Proceedings of the IEEE Workshop on Machine Learning for Signal Processing*. IEEE, Cancun, Mexico, 55–60. doi:10.1109/MLSP.2008.4685455

[23] Nordic Semiconductor. [n. d.]. Bluetooth Low Energy Fundamentals: Advertising Process. https://academy.nordicsemi.com/courses/bluetooth-low-energy-fundamentals/lessons/lesson-2-bluetooth-le-advertising/topic/advertising-process/

[24] Multiple open source contributors. 2018. Briar. GitLab. Available at: https://code.briarproject.org/briar/briar. Accessed on: 2025-11-03 21:00 GMT+0200.

[25] Aravindh Raman, Sagar Joglekar, Emiliano De Cristofaro, Nishanth Sastry, and Gareth Tyson. 2019. Challenges in the Decentralised Web: The Mastodon Case. In *Proceedings of the Internet Measurement Conference* (Amsterdam, Netherlands) *(IMC '19)*. Association for Computing Machinery, New York, NY, USA, 217–229. doi:10.1145/3355369.3355572

[26] Reliefweb. 2025. Gaza crisis: Update: Internet Connectivity Blackout in Gaza. Situation Report. Available at: https://reliefweb.int/report/occupied-palestinian-territory/gaza-crisis-update-internet-connectivity-blackout-gaza-16-june-2025-emergency-telecommunications-cluster-etc. Accessed on: 2025-11-03 21:00 GMT+0200.

[27] Tim Simonite. 2014. The Latest Chat App for iPhone Needs No Internet Connection. Article on MIT Technology Review. Available at: https://web.archive.org/web/20151127150930/http://www.technologyreview.com/news/525921/the-latest-chat-app-for-iphone-needs-no-internet-connection/. Accessed on: 2025-11-03 21:00 GMT+0200.

[28] Ion Stoica, Robert Morris, David Karger, M. Frans Kaashoek, and Hari Balakrishnan. 2001. Chord: A scalable peer-to-peer lookup service for internet applications. In *Proceedings of the 2001 Conference on Applications, Technologies, Architectures, and Protocols for Computer Communications* (San Diego, California, USA) *(SIGCOMM '01)*. Association for Computing Machinery, New York, NY, USA, 149–160. doi:10.1145/383059.383071

7 Appendix

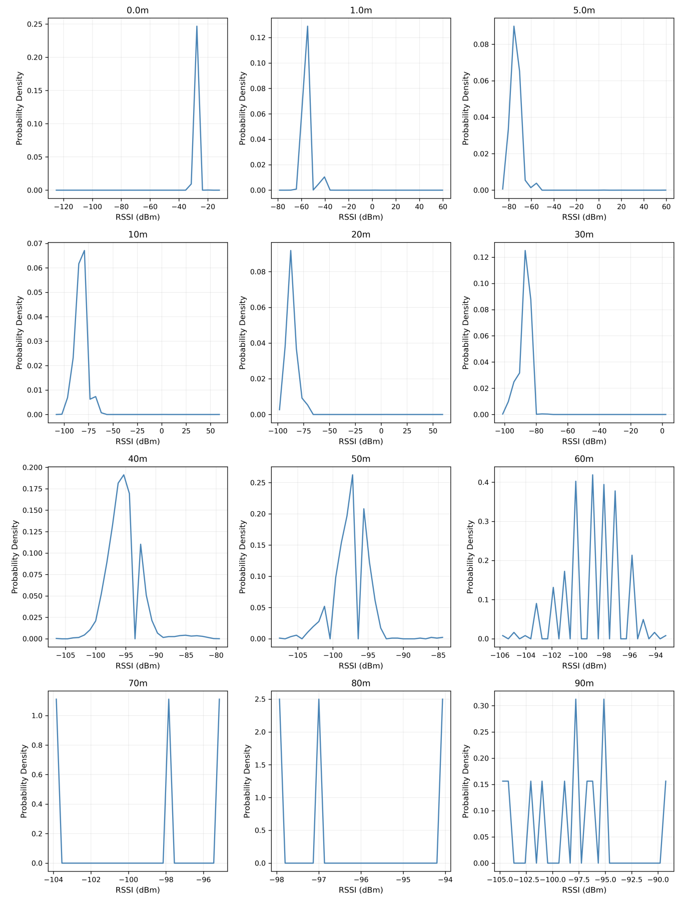

Figure 10: RSSI PDF

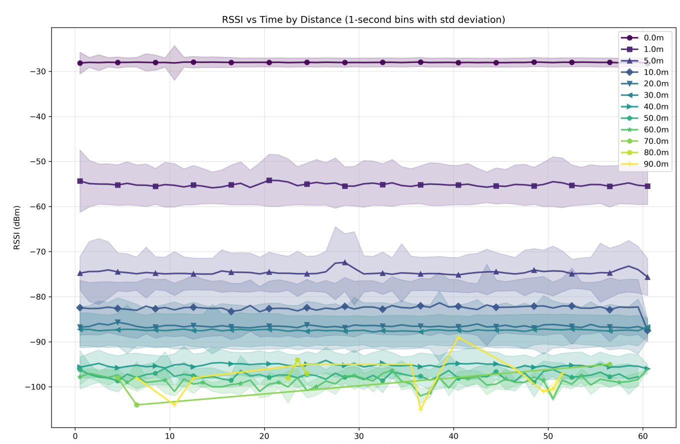

Figure 11: RSSI time series

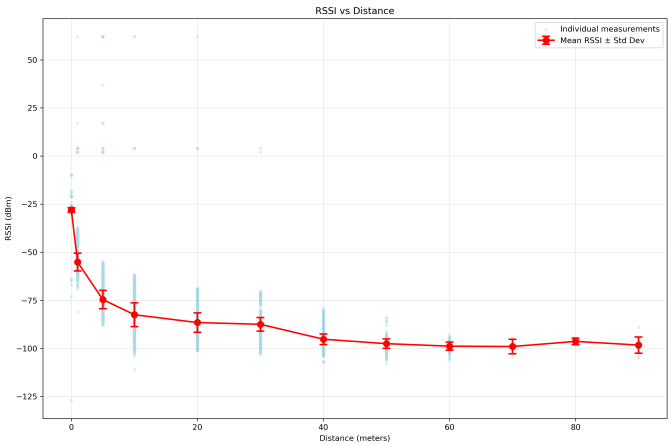

Figure 12: Mean RSSI at different distances

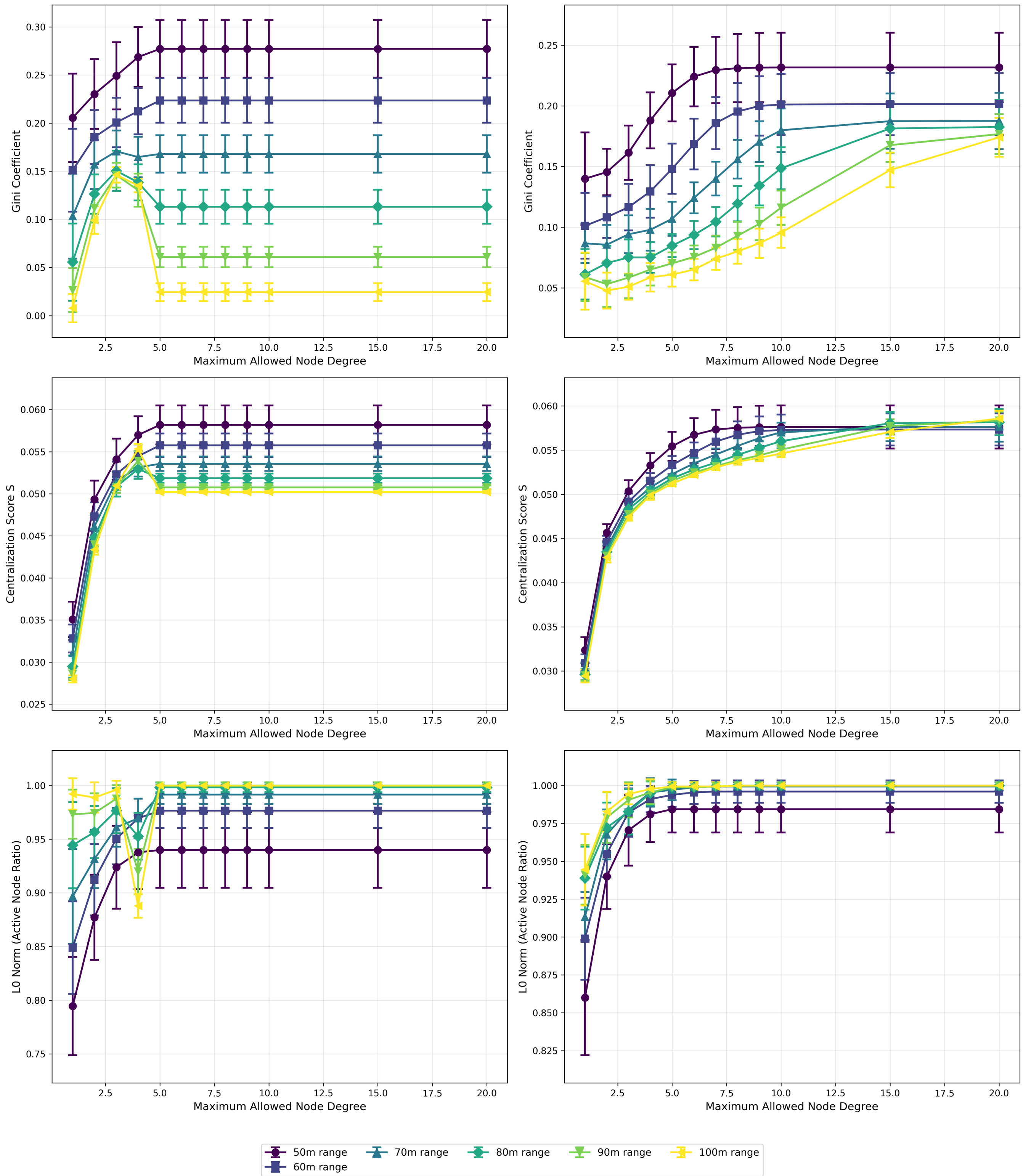

Figure 13: Centralization metrics vs maximum node degree in high communincation radius

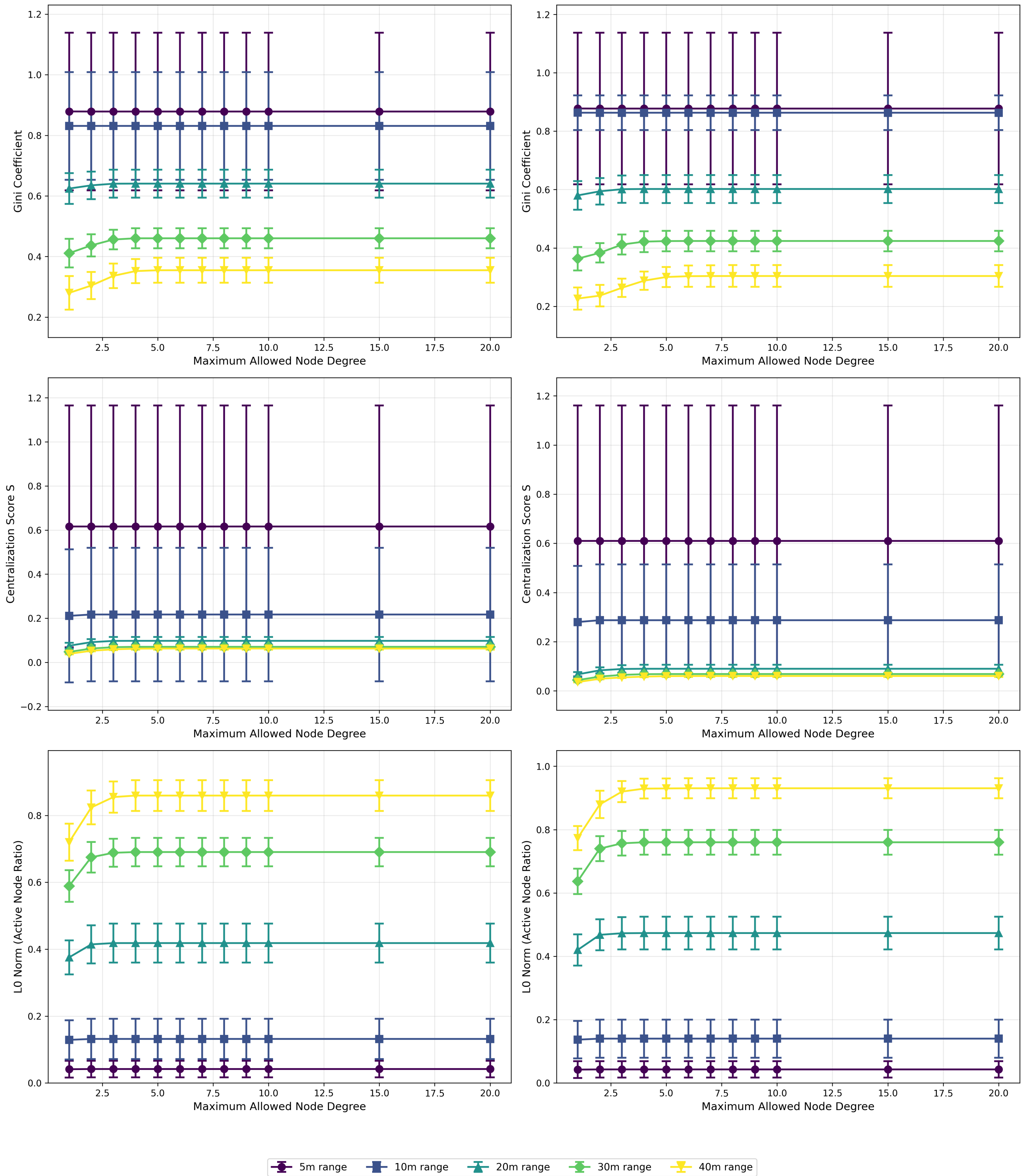

Figure 14: Centralization metrics vs maximum node degree in low communication radius